1. Create a virtual environment (once)

A **virtual environment** is a private box of packages for this project. It keeps the packages from clashing with other tools on your computer.

Open a terminal in this folder, then run:

macOS / Linux

```
python3 -m venv nlp-env
source nlp-env/bin/activate
```

Windows (Command Prompt)

```
python -m venv nlp-env
nlp-env\Scripts\activate
```

Windows (PowerShell)

```
python -m venv nlp-env
nlp-env\Scripts\Activate.ps1
```

When it is active, your prompt starts with `(nlp-env)`.

2. Install the packages

With the environment active:

```
pip install -r requirements.txt
pip install notebook ipykernel
```

> **Quotes matter.** If you type a single package with < or > yourself, put it in quotes: `pip install "transformers>=4.40,<5"`. Without quotes the terminal treats < and > as redirects and shows `command not found`. Using `-r requirements.txt` avoids the problem.

`transformers` must be **version 4** (<5). Version 5 removed some of the pipelines used in this lab.

3. Register the environment as a Jupyter kernel (once)

```
python -m ipykernel install --user --name nlp-env --display-name "Python (nlp-env)"
```

This makes `Python (nlp-env)` appear in Jupyter's kernel list.

4. Download the models (before class, with internet)

```
python download_models.py
```

- It downloads about **8 GB** in total. The biggest is the Nepali translation model (2.46 GB).
- If the connection drops, run the command again. Models that are already downloaded are not downloaded again.
- Models are saved in the Hugging Face cache and load offline afterwards:
  - macOS / Linux: `~/.cache/huggingface/hub`
  - Windows: `C:\Users\<your-name>\.cache\huggingface\hub`

5. Launch Jupyter Notebook

With the environment active, run:

```
jupyter notebook
```

Your browser opens. Open the lab notebook, then choose **Kernel > Change kernel > Python (nlp-env)** (in newer versions: **Kernel > Change Kernel** or the kernel picker at the top right).